

Real-time evacuation route optimization in the fire scenarios of cruise ships

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ABSTRACT

The optimization of evacuation routes for evacuees in fire scenarios on cruise ships directly affects evacuation efficiency. A time-varying equivalent weight Dijkstra algorithm (TVEWDA) is proposed, combining a fire dynamics simulator (FDS) to find optimal evacuation routes in case of a sudden cruise ship fire. First, the FDS software is used to simulate a cruise ship's single and multi-source fire scenarios. Second, the Dijkstra algorithm using only distance between nodes is improved by considering the effects of CO concentration, visibility, and temperature. The evacuees' speed is taken into consideration within the TVEWDA. The effects of congestion are considered in the evacuees' moving speed. Finally, the TVEWDA and a cruise ship evacuation network model are integrated to develop a real-time evacuation route optimization model. Simulation results show that the evacuation time achieved by the TVEWDA is 22% shorter than that of the Dijkstra algorithm using only distance between nodes and 11% shorter than that of an improved social force model in the multi-source fire scenario. This approach is well-suited to guide the evacuation of cruise ship evacuees in the event of single-source or multi-source fires, effectively ensuring the safety of passengers and the preservation of their property.

1. Introduction

Sea transportation plays a pivotal role in the global trade and transportation network, and modern cruise ships are remarkable in terms of large passenger capacity and long voyages [1,2]. In emergencies, evacuation is an effective means to ensure passenger safety. The confined environment of a sailing cruise ship and the emergency behaviors can affect the evacuation time when disasters occur [3,4]. Therefore, it is significant to study the optimization of evacuation routes in the event of a fire.

In recent years, many scholars have used surveys to examine the factors that influence evacuation. Arshad et al. [5] investigated the impact of passenger safety awareness on evacuation time through a questionnaire survey. Sarvari et al. [6] used a systematic approach to explore the literature on emergency evacuations from 1973 to 2017 and provided recommendations for evacuation management. Lee et al. [7] summarized the recent research advancements in passenger ship evacuation encompassing evacuee behavior, as well as evacuation optimization and evaluation. These studies serve as references for understanding evacuees' psychology, enhancing evacuation optimization methods, and improving safety management practices.

With the advancement of computer technology, simulation software is widely used in passenger ship evacuation analysis, including tools such as Pathfinder, AnyLogic, Fire Dynamics Simulator with Evacuation (FDS+EVAC), and Virtual Reality (VR) [8–11].

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Pathfinder software can be used to design evacuation plans and automatically generate navigation networks. Li et al. [12] introduced a method of passenger speed correction based on a fuzzy algorithm, employing Pathfinder software to verify the passenger evacuation speed. The AnyLogic software applies a social force model to simulate pedestrian movement, but it requires pre-set muster stations for evacuees. González-Cela et al. [13] proposed a multi-criteria decision framework based on AnyLogic, guiding the optimization of restaurant layouts on ships. Wang et al. [14] used FDS+EVAC to simulate emergency evacuations on passenger ships, offering practical recommendations for optimizing staircase layouts. VR technology can simulate three-dimensional environments, enabling an immersive experience for agents. Ginnis et al. [15] applied VR to study passenger ship evacuation scenarios, verified by the IMO/MSC 1033 advanced analysis method.

Choosing an optimal evacuation route is crucial for reducing evacuation time during emergencies. Various algorithms, such as the network flow algorithm, ant colony algorithm, and Dijkstra algorithm, are used for evacuation route optimization. Chu et al. [16] adopted the network flow algorithm to optimize fire evacuation routes on passenger ships, considering the impact of the fire at specific graph nodes. Wang et al. [17] proposed a cellular ant optimization model to address the path planning problem for evacuations on passenger ships. Łozowicka et al. [18] used a genetic algorithm to optimize evacuation routes on passenger ships. Liu et al. [19] applied an improved artificial fish swarming algorithm to dynamically plan evacuation routes on passenger ships while considering the effect of evacuation speed. These studies serve as references for passenger evacuation in emergencies. However, they do not consider the impact of fire risk factors on the evacuation process.

When planning evacuation routes for evacuees, it is essential to consider not only the impact of the fire, but also prioritize minimizing evacuation time [20]. To address the evacuation path problem in fire scenarios, numerous scholars have integrated FDS with path planning algorithms. Yan et al. [21] introduced an ant colony algorithm that accounts for the influence of fire smoke and uses a grid-based approach to build evacuation maps and enable the navigation of evacuees in the event of a fire in a subway station. Lee et al. [22] extracted the evacuation environment as a grid map and investigated the fire safety exit search method of the D* Lite algorithm. When applied to the complex and intricate evacuation environment of passenger ships, the grid map method becomes considerably challenging. On the other hand, the graph search method proves to be more stable and efficient compared to the previous grid map approach.

Several scholars have also used the graph search method to construct evacuation networks. Bernardini et al. [23] used FDS to model theater fires and designed an intelligent evacuation guidance system that monitors human behavior during evacuations, providing dynamic evacuation path recommendations. Cao et al. [24] considered the impact of both fire spread and distance when selecting safety exits for evacuation. Yang et al. [25] proposed a multi-objective algorithm for optimizing evacuation paths in subway fires and successfully applied it in Qingdao Metro. However, these methods primarily focus on buildings and do not account for intricate ship structures [26].

Fire evacuation on passenger ships is mainly simulation-based, and research combining FDS with path optimization algorithms is very limited [27]. Deng et al. [28] improved the A* algorithm to plan optimal evacuation paths for a 3-deck ferry in the event of a fire. Ji et al. [29] used the principles of a fuzzy neural network to evaluate comprehensive real-time situational risks and proposed an optimal path algorithm for fires on passenger ships. However, when a fire occurs on a passenger ship and disrupts the designated evacuation paths, managers face the challenge of re-planning the evacuation paths for all evacuees. This process can be inefficient due to frequent environmental changes during such incidents. In addition, the shortest evacuation path may not always be the safest [30, 31]. The real-time effects of CO concentration, visibility, and temperature during a passenger ship fire directly impact the evacuees [32]. To address these factors, the TVEWDA is proposed in this study, where the dynamic changes in evacuation speed during passenger ship fire evacuation are also considered, and the obtained optimal fire evacuation paths are imported into the simulation platform.

The Dijkstra algorithm is a common shortest route algorithm widely used for evacuation route optimization [33]. Li et al. [34] established an evacuation network diagram for passenger ships and implemented the Dijkstra algorithm to determine evacuation routes for evacuees. Mirahadi [35] defined risk factors by fire location and structural collapse and calculated the lowest risk route using a modified version of the Dijkstra algorithm. Zhu et al. [36] extended the Dijkstra algorithm from two-dimensional to three-dimensional space and predicted the evacuation routes for evacuees in the event of a flood. Darmi et al. [37] adopted the Dijkstra algorithm to determine the evacuation routes for evacuees in the event of an earthquake, effectively assisting the evacuation process. They also designed an information system to release the evacuation routes. Sari et al. [38] used the Dijkstra algorithm to find the shortest route from city A to city G. These studies utilized the Dijkstra algorithm to address route planning problems from a single departure point to a single destination.

In emergency situations, selecting an appropriate evacuation route plays a crucial role in enhancing evacuation efficiency. As a passenger ship fire spreads, each path in the evacuation network is affected to varying extents by the progression of the disaster. To address this issue, the concept of equivalent length is introduced, and fire-related factors are transformed into weights. Parameters such as CO concentration, visibility, and temperature are measured using FDS software. These parameters are then processed using Matlab calculations to determine the coefficients that reflect the influence of key risk factors. The resulting coefficients are recorded in a data sheet which is accessed by TVEWDA in this study. In the proposed TVEWDA, the objective function of the algorithm is to minimize the total evacuation time of all evacuees. This study introduces several innovations, including:

- (1) Real-time parameters of cruise ship fires were simulated using FDS, and the resulting fire simulation data were then input into the dynamic evacuation route optimization model.

- (2) The TVEWDA is proposed considering the hazards associated with the spread of fire. Parameters such as CO concentration, visibility, and temperature are converted into equivalent lengths. Additionally, the effects of congestion are considered in the evacuees' movement speed.
- (3) The dynamic route optimization model for passenger evacuation is applied to the cruise ship evacuation system in scenarios involving single-source and multi-source fires, providing guidance on evacuation routes for evacuees in the event of a fire on a passenger ship.

This study is organized as follows. Section 2 analyzes the FDS simulation, establishes the evacuation network, and proposes the TVEWDA. Section 3 presents the results of the cruise ship's single and multi-source fire scenarios. Finally, Section 4 concludes this work by highlighting the key findings and discussing certain limitations.

2. Methodology

2.1. Establishment of evacuation network

The research subject is a 10-deck cruise ship. The cruise ship has a dining room and cabins on the fifth deck, and it is one of the key locations to fire prevention. The fifth deck of the cruise ship is 368 m long, 32 m wide, and 3 m high. According to the cruise ship CAD diagram, the node network graph $G(V, E, W)$ is established for the fifth deck. Here, V represents the nodes that indicate reachable locations for evacuees on the cruise ship, typically including cabins, corridors, entertainment rooms, and other accessible areas, E represents the feasible routes between connecting nodes, and W represents the weight, which usually refers to the length of the evacuation route. The cruise ship network model is established according to the following two principles: (1) Nodes are placed at locations that evacuees can reach. (2) The distance between each node in the evacuation network can be calculated.

Fig. 1 illustrates the layout of the fifth deck, and Fig. 2 represents its node network graph. Fig. 3 highlights the specific locations of nodes 1~69, while Fig. 4 shows the specific locations of nodes 70~117. Notably, the fifth deck has 117 nodes. Nodes [16, 18, 27, 41, 42, 58, 70, 72, 89, 91, 114] correspond to stairs, and the remaining nodes are locations evacuees can reach, such as corners, passages, and cabins.

2.2. FDS simulation

Many studies have shown that high fire risk locations include cabins, dining rooms, engine rooms, and others. Since the fifth deck of



Fig. 1. Layout of the fifth deck.

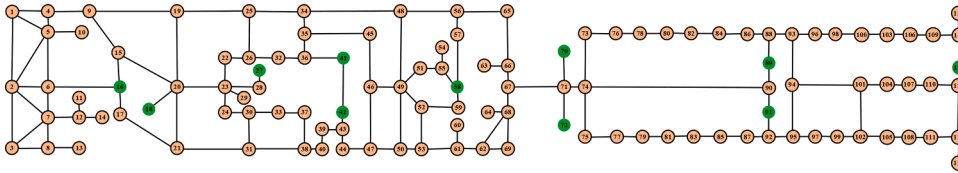


Fig. 2. Network graph of the fifth deck.

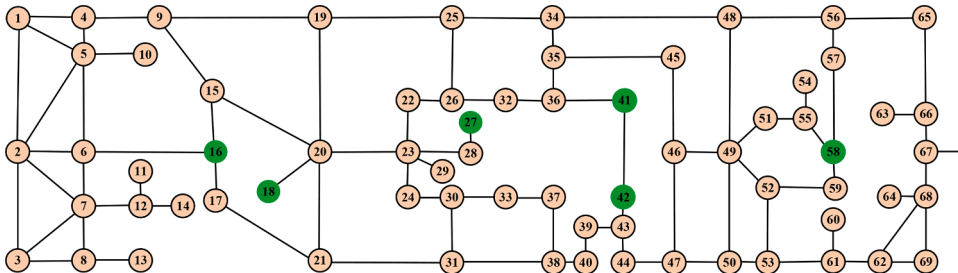


Fig. 3. Specific locations of nodes 1~69.

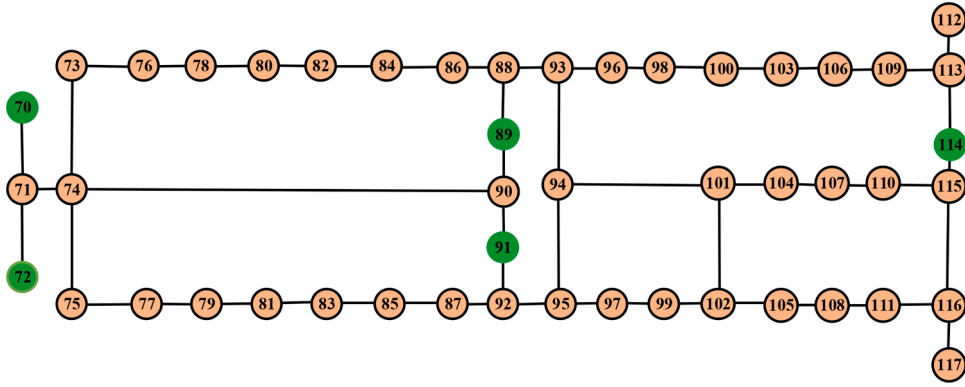


Fig. 4. Specific locations of nodes 70~117.



Fig. 5. Fire simulation model of the fifth deck.

the cruise ship contains cabins, a dining room, and an entertainment room, it is chosen as the focus of this research to build a fire simulation model, as shown in Fig. 5.

PyroSim was developed by the National Institute of Standards and Technology in the US. It is a fire simulation software based on an FDS [39]. In this work, fire dynamics are simulated using PyroSim. The accuracy of simulations involving buoyant plumes relies on both the size of the fire and the size of the mesh cells. The characteristic fire diameter is determined according to the guidelines provided in the FDS user guide [40]:

$$d^* = \left(\frac{Q}{\rho_0 c_p T_0 g^{1/2}} \right)^{2/5}, \quad (1)$$

$$4 \leq \frac{d^*}{\psi_x} \leq 16, \quad (2)$$

where d^* is the characteristic diameter of the fire (m), ψ_x is the size of the mesh cell (m), Q is the total heat released by the fire (kW), ρ_0 is the air density (kg/m^3), c_p is the specific heat capacity of the air ($\text{kJ}/\text{kg} \cdot \text{K}$), T_0 is the ambient temperature (K), and g is the gravitational acceleration (m/s^2).

Relatively small values of d^* and/or large grid cells are suitable for estimating the flame height and structure [41]. According to the IMO test protocol, the total heat released Q is 6.0 Mw, which is typical in ship fires [42]. Considering the meteorological characteristics of passenger ships navigating the South China Sea, the simulation is conducted for passenger ships during the summer season. The air density ρ_0 is $1.169 \text{ kg}/\text{m}^3$, the specific heat capacity c_p of the air is $1.004 \text{ kJ}/\text{kg} \cdot \text{K}$, and the ambient temperature T_0 is 20°C [43]. Based on these parameters, the value of d^* is calculated to be 1.99 m.

To satisfy the conditions of (2), ψ_x should vary between 0.2 m and 0.8 m [44]. Therefore, a relatively coarse mesh ($0.8 \text{ m} \times 0.8 \text{ m} \times 0.8 \text{ m}$) was initially created and gradually refined until no significant differences were observed in the results. The grid size of the mesh was set to $0.5 \text{ m} \times 0.5 \text{ m} \times 0.5 \text{ m}$. Therefore, the PyroSim software automatically determined that the mesh of the fifth deck is divided into $489 \times 78 \times 6$ grid cells, resulting in a total of 228,852 mesh cells. The properties of the materials mainly used in the cruise ship fire simulation are presented in Table 1.

Usually, the combustion model of cruise ship fires is a non-steady state model. In this study, the same model is adopted as the fire simulation model. The mathematical expression of the model is:

$$\dot{Q} = \alpha_r t^2, \quad (3)$$

where $\dot{Q}$ is the heat release rate per unit time at the source of fire, α_r is the growth coefficient of the model (kW/s^2), and t is the time (s). The relationship between α_r and the type of fire growth is shown in Table 2 [45].

The heat and smoke generated by a fire can impair the judgment of evacuees. In this study, CO concentration, visibility, and temperature are considered as key risk factors. The International Maritime Organization has proposed four performance criteria for ensuring life safety, as shown in Table 3 [46].

The calculation equation for the influence coefficient of the evacuation route between nodes i and j is given by:

Table 1
Properties of the materials.

Parameter	Steel	Aluminum	Board	Gypsum
Emissivity (-)	0.8	0.55	0.92	0.8
Density (kg/m ³)	7850	2670	500	1050
Specific heat capacity (kJ/kg · K)	0.48	0.8	2.5	1.0
Heat conductivity (W/m · K)	58.2	237	0.35	330

$$q_{ij}^g = \frac{mgv_0^a \sin \theta_{ij}}{R_0} + \cos \theta_{ij}, \quad (4)$$

where m is the mass of the human body (kg), with the average weight of evacuees $m = 64.3$ kg, v_0^a is the average initial velocity of evacuees, with the average evacuation speed $v_0^a = 1.1$ m/s, g is the acceleration of gravity (m/s^2), θ_{ij} is the angle of inclination, with $\theta_{ij} = 0^\circ$ for horizontal corridors and $\theta_{ij} = 45^\circ$ for stairways, and R_0 is the mobility (W), with the average mobility $R_0 = 200$ W.

The calculation equations for the influence coefficients of key risk factors between nodes i and j are given in (5)-(8):

$$q_{ij}^v = \frac{1}{f_1(t)} + \frac{1}{f_2(\alpha)} + \frac{1}{f_3(\rho)} - 2, \quad (5)$$

where $f_1(t)$ is the temperature influencing factor, $f_2(\alpha)$ is the visibility influencing factor, and $f_3(\rho)$ is the CO concentration influencing factor.

Zhu et al. [47] studied the effect of temperature on the evacuation speed of passengers during a fire on a passenger ship and derived a formula for the temperature influencing factor:

$$f_1(t) = \begin{cases} 1, & (t_e \leq 30^\circ\text{C}) \\ \frac{(v_{\max} - 1.2) \left(\frac{t_e - t_{c1}}{t_{c2} - t_{c1}} \right)^2}{1.2} + 1, & (30^\circ\text{C} < t_e \leq 60^\circ\text{C}) \\ \frac{v_{\max}}{1.2} \left[1 - \left(\frac{t_e - t_{c2}}{t_{dead} - t_{c2}} \right)^2 \right], & (60^\circ\text{C} < t_e \leq 120^\circ\text{C}) \\ 0, & (t_e > 120^\circ\text{C}) \end{cases} \quad (6)$$

where $f_1(t)$ denotes the influence of temperature, t_e indicates the ambient temperature, t_{c1} is the temperature at which the evacuees feel uncomfortable ($t_{c1} = 30^\circ\text{C}$), t_{c2} is the temperature at which the evacuees are injured ($t_{c2} = 60^\circ\text{C}$), t_{dead} is the temperature which causes the death of the evacuees ($t_{dead} = 120^\circ\text{C}$), and $v_{\max}$ is the maximum evacuation speed ($v_{\max} = 4$ m/s).

To study the effect of visibility during a passenger ship fire, several experiments were conducted and an equation for the factors affecting visibility was established as follows [48]:

$$f_2(\alpha) = \max \left\{ 0.1, 1 - \frac{0.057}{0.706} \alpha \right\}, \quad (7)$$

where $f_2(\alpha)$ indicates the influence of visibility and α denotes the extinction coefficient of fire smoke (m^{-1}).

Milke et al. [49] studied the influence factor of CO concentration as shown in (8):

$$f_3(\rho) = \begin{cases} 1, & \rho < 0.1 \\ 1 - (0.2125 + 1.788\rho)\rho t, & 0.1 \leq \rho < 0.25 \\ 0, & \rho \geq 0.25 \end{cases} \quad (8)$$

where $f_3(\rho)$ represents the influence factor of CO concentration, ρ is the volume of CO, and t is the exposure time.

In this study, fire product data were recorded at each evacuation node during the fire event by installing CO concentration, temperature, and visibility detectors. The detectors were positioned at a height of 1.8 m above the ground.

Table 2
Relationship between fire coefficient and fire type.

Fire type	Fire coefficient (kW / s ²)
Slow	0.0029
Medium	0.0117
Fast	0.0469
Super fast	0.1876

Table 3
Classification of key risk factors.

Hazard factors	Critical values
Temperature (°C)	60
Visibility (m)	10
CO (ppm)	100

2.3. TVEWDA

In the event of a fire, the evacuation objective for the evacuees is to minimize the evacuation time. To achieve this goal, the TVEWDA is proposed for passenger evacuation on cruise ships. It can be applied to scenarios with single-source and multiple-source fires.

The Euclidean distance between nodes i and j in the network graph is expressed as:

$$l_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}, \quad (9)$$

where l_{ij} is the Euclidean distance (m), x_i is the horizontal coordinate of node i , and y_i is its vertical coordinate.

The evacuation routes during a passenger ship fire are divided into two categories: feasible and emergency evacuation routes. Feasible evacuation routes are paths where the risk factors remain within a safe range. Emergency evacuation routes, on the other hand, are paths where the fire risk factor threshold exceeds the safety standard, and failure to escape in time could jeopardize the evacuees' lives. Wang [50] concluded that the evacuation speed of evacuees on feasible evacuation routes is 1 m/s, and it increases on emergency evacuation routes. Tsunemine et al. [51] showed that the speed of evacuees is 1 m/s under normal conditions, and 1.8 m/s when the fire is present. Considering that when developing evacuation routes, it is important to ensure that all evacuees evacuate safely, this study assumes that evacuees evacuate 1.8 times faster than the feasible evacuation routes when they are on the emergency evacuation routes. During the evacuation process, crowd congestion can have a significant impact on evacuation speed. The speed at which evacuees move between nodes i and j is determined by:

$$v_{ij}(t) = v_0 \eta e^{-\lambda t}, \quad (10)$$

where λ is the crowding factor and η is the emergency behavior factor, which is 1 for feasible evacuation routes between nodes i and j , and 1.8 for emergency evacuation routes.

Crowding during evacuation has a significant impact on evacuation efficiency [52]. The crowding factor λ is defined as the ratio of the number of passengers on edges i and j at time t to the equivalent distance of the edge:

$$\lambda = \frac{N_{ij}(t)}{L_{ij}}, \quad (11)$$

where $N_{ij}(t)$ is the number of passengers on edges i and j at time t .

In consideration of the key risk factors, the equivalent distance between nodes i and j is calculated as:

$$L_{ij} = q_{ij}^g q_{ij}^v l_{ij}, \quad (12)$$

where L_{ij} is the equivalent length of the path between nodes i and j , q_{ij}^g is the influence coefficient of the evacuation route and q_{ij}^v is the influence factor of fire.

The evacuation time for passengers moving from node i to node j is given by:

$$t_{ij} = \frac{L_{ij}}{v_{ij}}, \quad (13)$$

When evacuees start to evacuate from the cruise ship, they are scattered all over the ship [53]. These evacuees independently determine their paths from the starting point to the destination, ensuring minimal travel time between nodes. The time for an evacuee to evacuate depends on the evacuation path, denoted as P , which constitutes a collection of nodes spanning from the starting point to the destination. The evacuation path P is defined as follows:

$$P = \{P_1, \dots, P_n\}, \quad (14)$$

where P is the path for each evacuee, and $P_1, \dots, P_n$ are the nodes of P .

In detail, if $t_{P_i, P_{i+1}}$ is the travel time to move between nodes i and $i + 1$, the evacuation time $T(P)$ on path P is expressed as:

$$T(P) = \sum_{i=1}^{n-1} t_{P_i, P_{i+1}}, \quad (15)$$

where $T(P)$ is the evacuation time on path P .

This study employs the TVEWDA to search for the optimal path for each evacuee. This algorithm assigns to each evacuee k the minimum evacuation time T_k . Namely, if $\{P\}_k$ is the collection of all possible paths available to k , the minimum evacuation time T_k is defined as follows:

$$T_k = \min_{P \in \{P\}_k} T(P), \quad (16)$$

Finally, the evacuation performance is evaluated using the longest evacuation time $T_{\max}$ between all evacuees, which can be calculated as:

$$T_{\max} = \max_k T_k, \quad (17)$$

where $T_{\max}$ is the longest evacuation time between all evacuees.

The TVEWDA is proposed based on the Dijkstra algorithm. The main flowchart illustrating the process of crowd evacuation using the TVEWDA is shown in Fig. 6:

The main steps of the TVEWDA are as follows:

- (1) Draw a crowd evacuation node diagram based on the cruise ship CAD drawings, noting the coordinates (x_i, y_i) of each node and the length of each route l_{ij} .
- (2) Use FDS simulation to determine the influence of fire smoke temperature $f_1(t)$, visibility $f_2(\alpha)$, and CO concentration $f_3(\rho)$. Calculate the equivalent length of evacuees running between nodes i and j .
- (3) Establish the adjacency matrix, edge capacity matrix, and evacuees' distribution matrix for the evacuation network nodes.

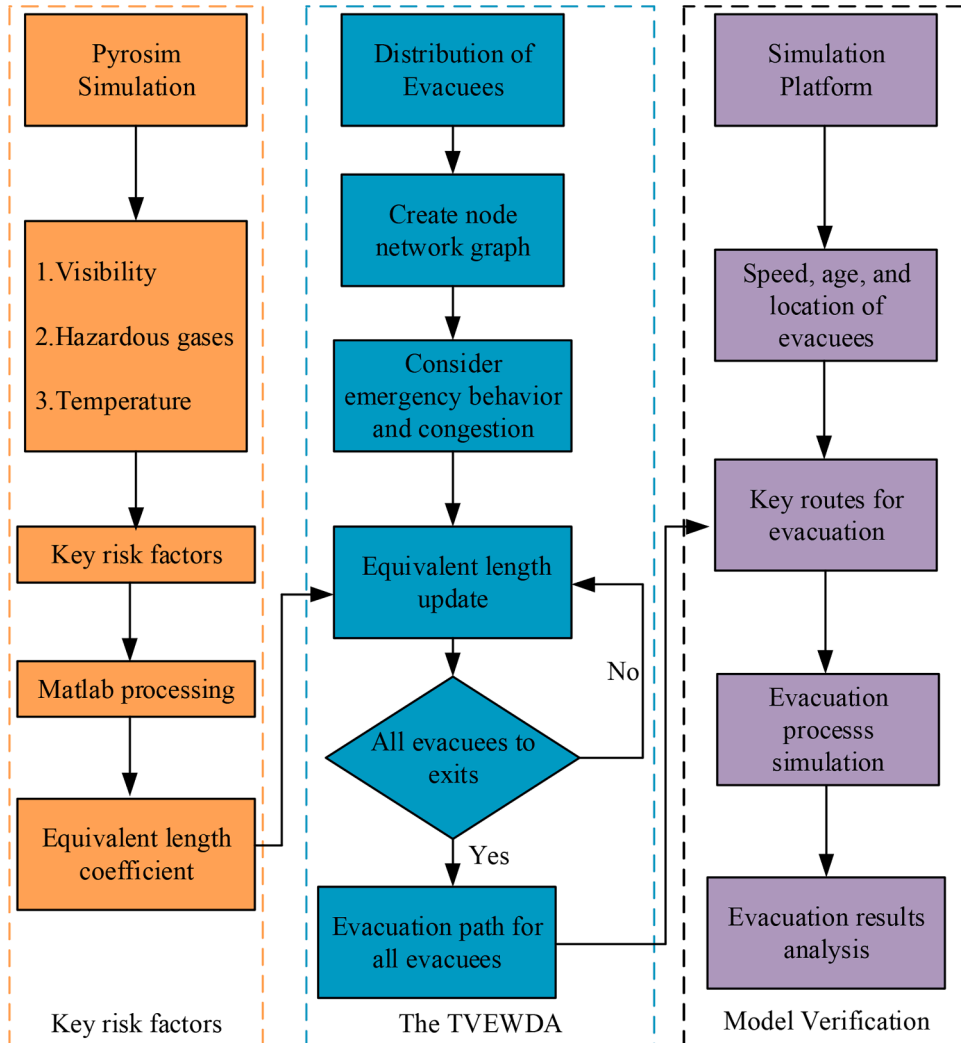


Fig. 6. The flowchart of the TVEWDA.

- (4) Apply the TVEWDA to search for optimal evacuation route for all evacuees, taking into account the shortest time for reaching the stairs, while considering the effects of emergency behavior and congestion.
- (5) Check if all evacuees have reached the stairs; consider the evacuation is complete in this case and calculate the equivalent length and evacuation time of evacuees. Otherwise, return to Step 2.
- (6) Input the evacuation routes into the simulation platform to assess the effectiveness of the algorithm.

3. Results and discussion

3.1. Impact of single-source fire on evacuation route

The dining room is a central place for human activities. In the event of a fire, it may spread rapidly within the dining room. Failure to promptly and effectively evacuate passengers can result in casualties. Therefore, in this research, the fire was deliberately located in the dining room. Detectors were set up to monitor the CO concentration, visibility, and temperature at different times during the fire, and Figs. 7–9 show snapshots of these measurements at each time.

Fig. 7 shows that when a fire occurs, the CO concentration rapidly exceeds the critical threshold in the dining room. Fig. 8 and Fig. 9 demonstrate similar trends in the changes of visibility for evacuees and the temperature near the fire. Increased CO concentrations, reduced visibility, and elevated temperatures directly hinder the evacuation speed of individuals. In the event of a fire, inhaling water vapor-saturated air with temperatures above 60 °C can cause thermal burns to the human body. This study sets the critical risk factors for evacuation at 100 ppm for CO concentration, 10 m for visibility, and 60 °C for temperature. Although the loss of visibility during a fire may not directly impact the vital signs of evacuees, it can induce emergency behavior and unease among them, potentially leading to a stampede accident. Therefore, the visibility threshold of 10 m is set.

To evaluate the risk factor values at the critical location during a fire, the temperature, visibility, and CO concentration values obtained by the installed detectors are shown in Fig. 10, Fig. 11, and Fig. 12, respectively.

As shown in Fig. 10, when a fire occurred on the cruise ship, the temperature at node 27 exceeded 60 °C in about 90 s, while the

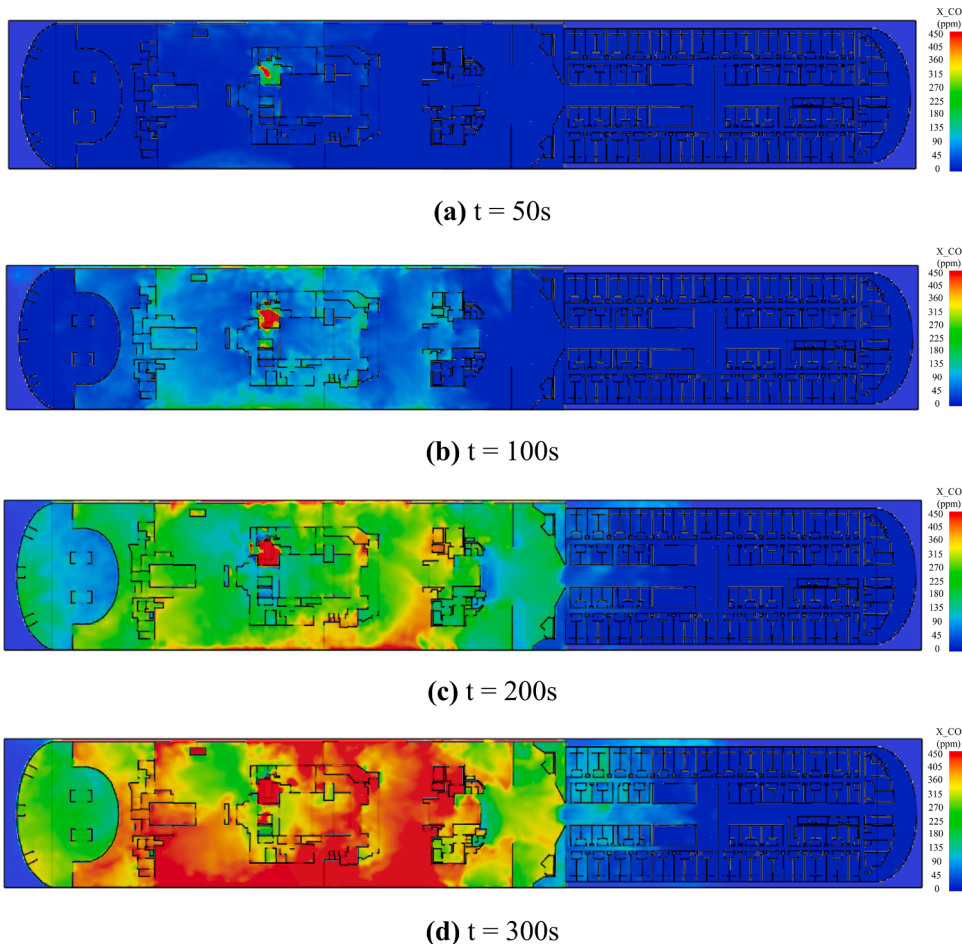


Fig. 7. CO concentration at different times of the fire.

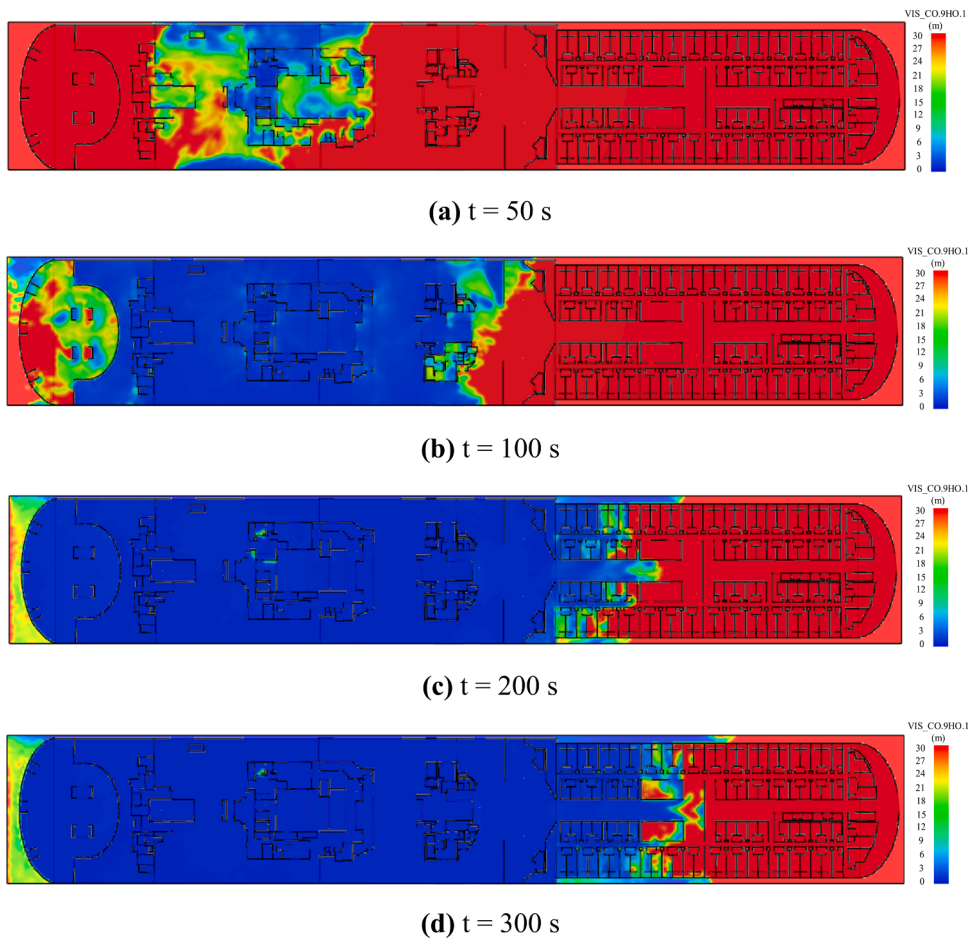


Fig. 8. Visibility at different times of the fire.

temperature at nodes 16 and 58 did not significantly change due to their distance from the fire source. As shown in Fig. 11, during the fire incident, the visibility dropped below 10 m at nodes 27, 41, and 42 in roughly 60.0 s, 47.7 s, and 52.5 s, respectively. As depicted in Fig. 12, the CO concentration at nodes 27 and 18 is 100 ppm in about 76.5 s and 102.0 s, respectively. At node 41, CO concentrations reaches 100 ppm in approximately 81 s, and at node 42, it reaches 103 ppm in about 102 s.

The evacuation drill simulated the process of evacuating under extreme conditions, such as fires outbreak in the dining area and cabins. A total of 600 evacuees were involved on the fifth deck of the cruise ship. Out of these, 265 were designated to evacuate from the cabins on the fifth deck, while the remaining 335 were dispersed across various areas like the supermarket, dining hall, and entertainment rooms. The participants encompassed a diverse demographic: 162 adult males, 150 adult females, 100 elderly males, 140 elderly females, and 48 children. The distribution of evacuees is based on the evacuation drill, in which a total of 600 evacuees participated. Adult males accounted for 27%, adult females accounted for 25%, the elderly accounted for 40%, and children accounted for 8%. In this study, adult males were observed to evacuate at a speed of 1.19 m/s, adult females at a speed of 0.98 m/s, the elderly at a speed of 0.55 m/s, and children at a speed of 0.64 m/s. Fig. 13 shows the specific locations of evacuees' distribution: 20 evacuees were distributed at the sightseeing desk, 315 evacuees in the dining and entertainment rooms, and 265 evacuees in the cabins.

When a fire broke out at node 22, the evacuation route for several evacuees at the early stage of the fire in about 50 s is shown in Table 4.

When a fire occurs at node 22, the evacuation route for evacuees is shown in Fig. 14(a). As the fire spreads from node 22 to nodes 23, 26, 27, and 28, the network diagram for the evacuation of evacuees from adjacent nodes is shown in Fig. 14(b).

As shown in Fig. 14(a), when a fire occurred at node 22, the routes connecting node 22 with nodes 23 and 26 were inaccessible due to CO concentration, high temperature, and smoke generated by the fire. Therefore, the optimal route for node 26 was changed from 26→22→23→28→27 to 26→32→36→41. The optimal evacuation route for node 25 was 25→19→20→18, and the optimal evacuation route for node 24 was 24→30→33→37→38→40→39→43→42. As shown in Fig. 14(b), when the fire spread to nodes 22, 23, 26, 27, 28, and 29, the routes between nodes (22, 23), (22, 26), (23, 29), (23, 28) and (27, 28) were inaccessible. The optimal route for node 25 was 25→19→20→18. The optimal evacuation route for node 24 was 24→30→33→37→38→40→39→43→42.

In case of a fire, the shortest evacuation route is shown in Fig. 15 for comparison.

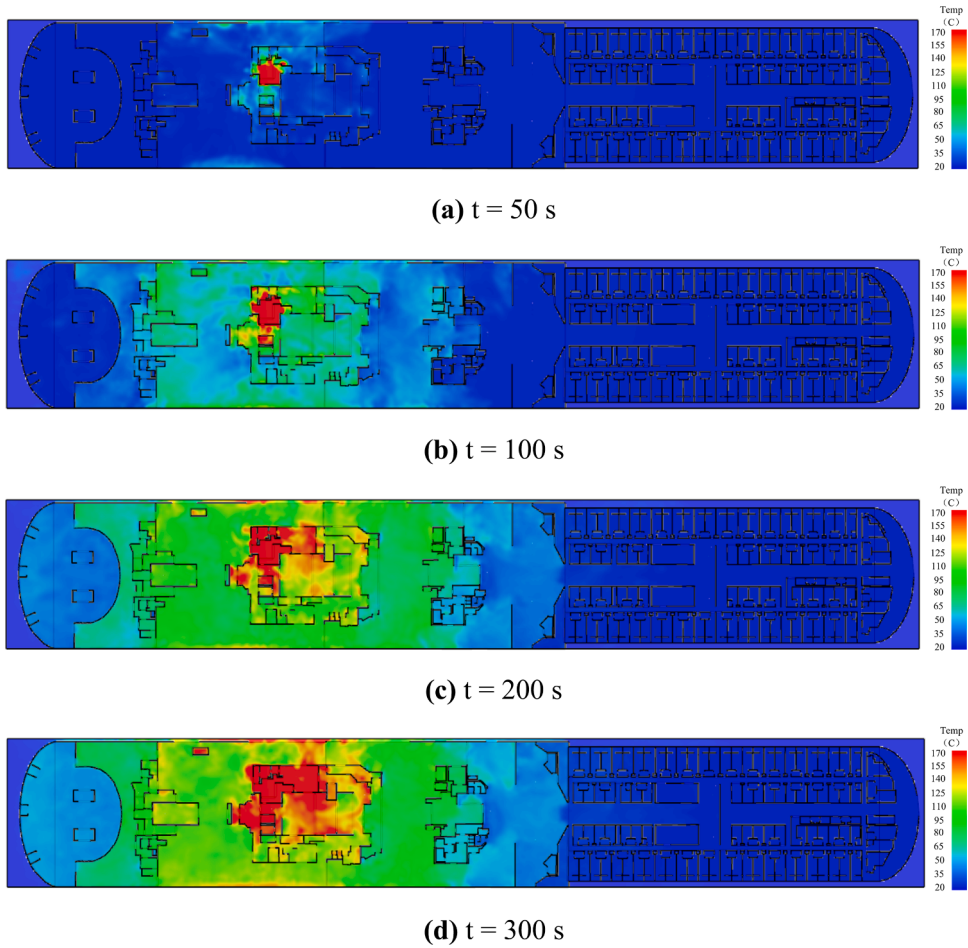


Fig. 9. Temperature at different times of the fire.

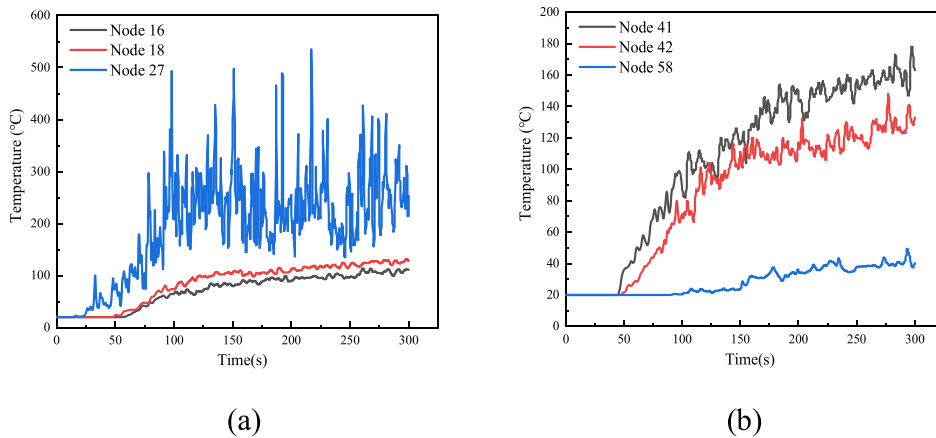


Fig. 10. Change in temperature at key locations.

As shown in Fig. 14(a) and Fig. 15, the TVEWDA and the Dijkstra algorithm using only distance between nodes differ significantly in the evacuation routes for nodes 25 and 24. When a fire broke out, according to the Dijkstra algorithm using only distance between nodes, the evacuation route for node 25 was 25→26→32→36→41, and the evacuation route for node 24 was 24→23→28→27. Considering the influence of key risk factors in the event of a fire, the TVEWDA calculated the evacuation route for node 25 as 25→19→20→18, and the evacuation route for node 24 as 24→30→33→37→38→40→39→43→42.

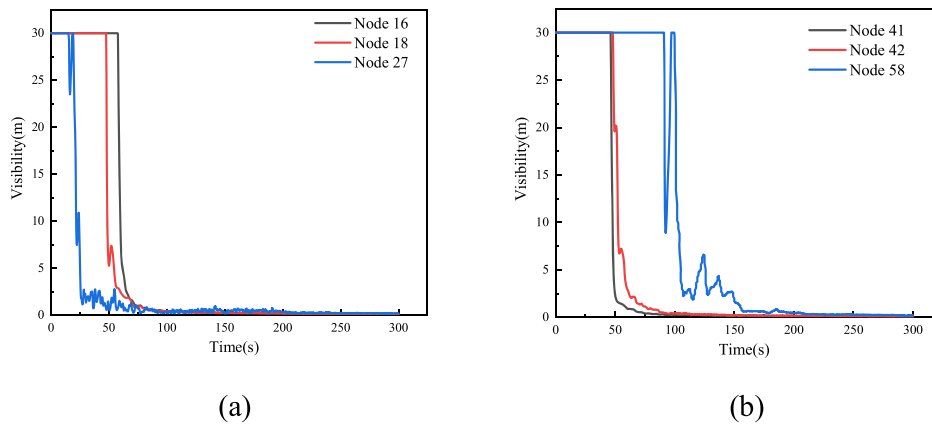


Fig. 11. Change in visibility at key locations.

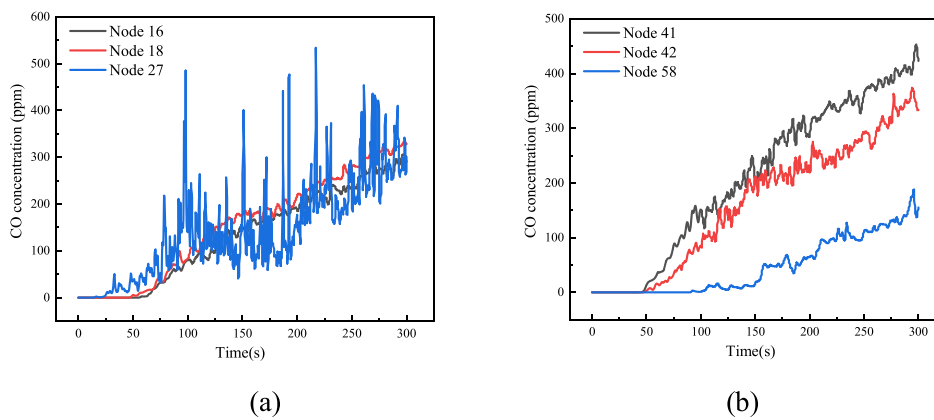


Fig. 12. Change in CO concentration at key locations.



Fig. 13. Specific location of evacuees' distribution.

Table 4

Evacuation route of several evacuees ($t = 50$ s).

Start	End	Optimal route of the TVEWDA	Time (s)
1	16	1→2→6→16	51.9
4	16	4→9→15→16	41.5
14	16	14→12→7→6→16	42.0
23	27	23→28→27	13.4
46	42	46→47→44→43→42	58.3
80	89	80→82→84→86→88→89	65.3
81	91	81→83→85→87→92→91	66.2
100	114	100→103→106→109→113→114	73.2
103	114	103→106→109→113→114	50.0
105	114	105→108→111→116→115→114	60.7

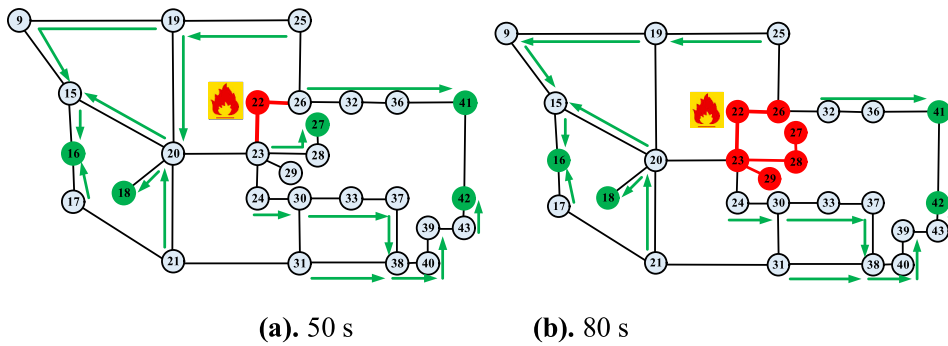


Fig. 14. Evacuation route at different times (the TVEWDA).

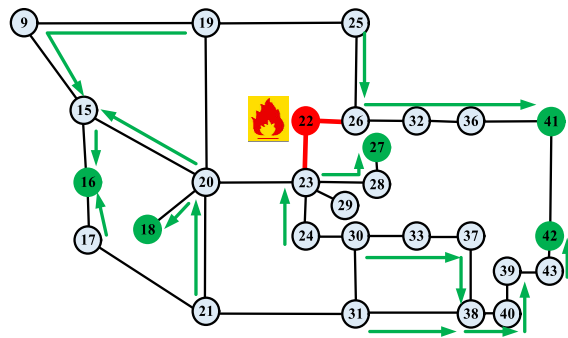


Fig. 15. Evacuation route in about 50 s (the Dijkstra algorithm).

The evacuation time curves for the TVEWDA and the Dijkstra algorithm using only distance between nodes are shown in Fig. 16.

As shown in Fig. 16, the red curve represents the evacuation curve of the Dijkstra algorithm using only distance between nodes, and the black curve is the evacuation curve of the TVEWDA. The evacuation time of the TVEWDA is 21% shorter than that of the Dijkstra algorithm using only distance between nodes. Because TVEWDA considers the emergency behavior and congestion during passenger ship evacuation, it results in a faster evacuation time compared to the Dijkstra algorithm using only distance between nodes.

3.2. Impact of multi-source fires on evacuation route

This study also simulated extreme conditions of simultaneous fires in the dining room and cabin. Fig. 17, Fig. 18, and Fig. 19 present snapshots of the CO concentration, visibility, and temperature in the event of a multi-source fire, respectively.

Fig. 17, Fig. 18, and Fig. 19, respectively show the variations in CO concentration, visibility, and temperature at different time intervals during a multi-source fire. In the figures, the left fire source represents the fire in the dining room, while the right fire source represents the fire in the cabin. Fig. 17 demonstrates that at $t = 100$ s, the CO concentration at the ignition point in the cabin exceeded 100 ppm. As observed in Fig. 18, visibility at certain key locations drops below 10 m within approximately 100 s, potentially causing emergency behavior and unease among evacuees. Fig. 19 shows that when $t = 6$ s, the dining room where the fire originated becomes a high-temperature zone with heat spreading to other areas. By $t = 120$ s, the temperature of the cabin continues to rise, approaching the maximum temperature that the human body can tolerate, i.e., 60°C . At $t = 130$ s, harmful gasses spread throughout the dining room, posing significant challenges to evacuation.

To examine the risk factor values at critical locations during a fire, the values of temperature, visibility, and CO concentration obtained by the installed detectors are respectively shown in Fig. 20, Fig. 21, and Fig. 22.

As shown in Fig. 20, when a fire broke out on the cruise ship, the temperature at nodes 27 and 41 exceeded 60°C in about 75 s and 163 s, respectively. The temperature changes at nodes 91 and 114 were not significant. As shown in Fig. 21, the visibility of evacuees under normal conditions was 30 m. However, when a fire occurred, the visibility at nodes 27 and 41, near the fire source, dropped below 10 m in about 32 s and 54 s, respectively. Fig. 22 shows that the concentration of CO near the fire point rapidly exceeded the critical threshold. At $t = 126$ s, the CO concentration at node 41 exceeded 100 ppm.

After 50 s of the fire's initiation, nodes 22 and 89 near the fire source became inaccessible. Consequently, the evacuation routes for certain evacuees had to be altered, as shown in Table 5.

As shown in Table 5, the evacuation routes for nearby nodes were changed due to the inaccessibility of nodes 22 and 89. The optimal route for node 96 changed from $96 \rightarrow 93 \rightarrow 88 \rightarrow 89$ to $96 \rightarrow 93 \rightarrow 94 \rightarrow 95 \rightarrow 92 \rightarrow 91$.

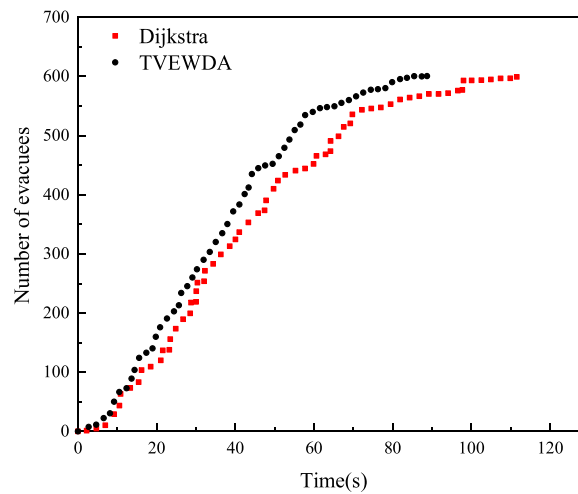


Fig. 16. Comparison of evacuation curves for different algorithms.

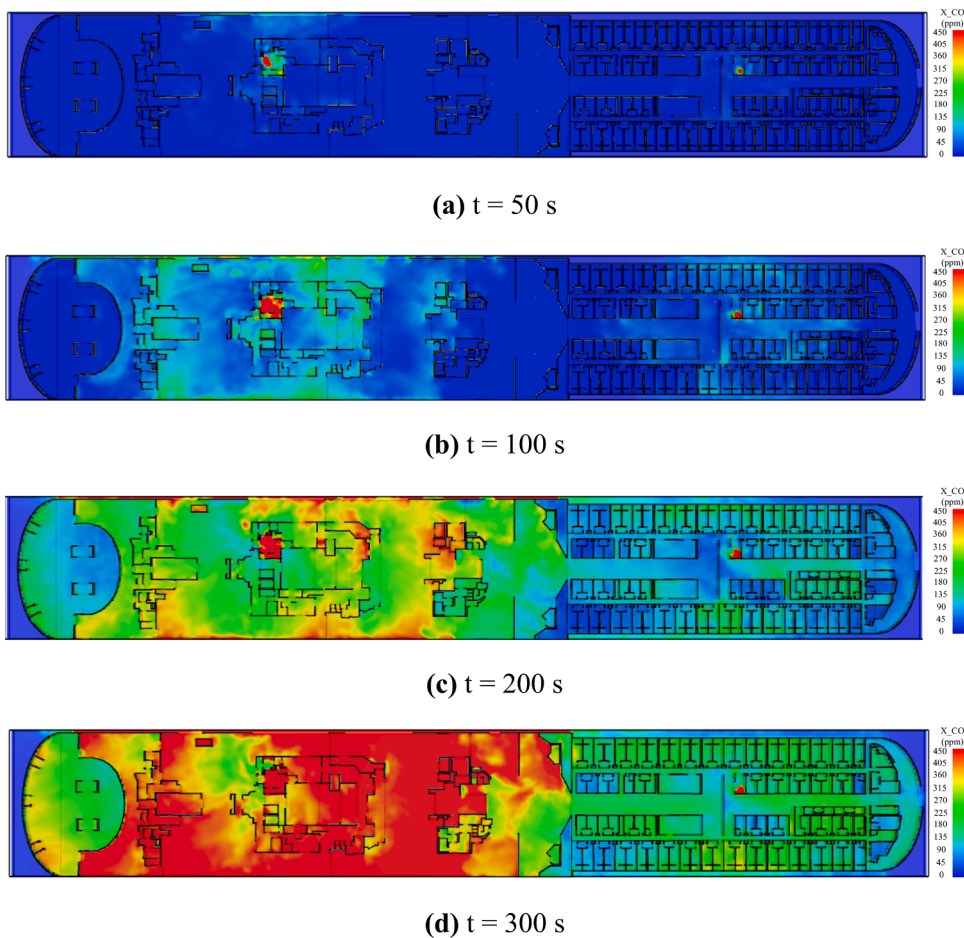


Fig. 17. CO concentration at different times in case of a multi-source fire.

After 100 s of the fire's initiation, the nodes in close proximity to the fire source, including nodes 22, 23, 26, 27, 28, 29, 88, 89, 93, and 96, became impassable. Consequently, the evacuation routes for some evacuees had to be adjusted, as shown in Table 6.

As shown in Table 6, since nodes 22, 23, 26, 27 and 28 became inaccessible, their evacuation routes were altered as in the case of a

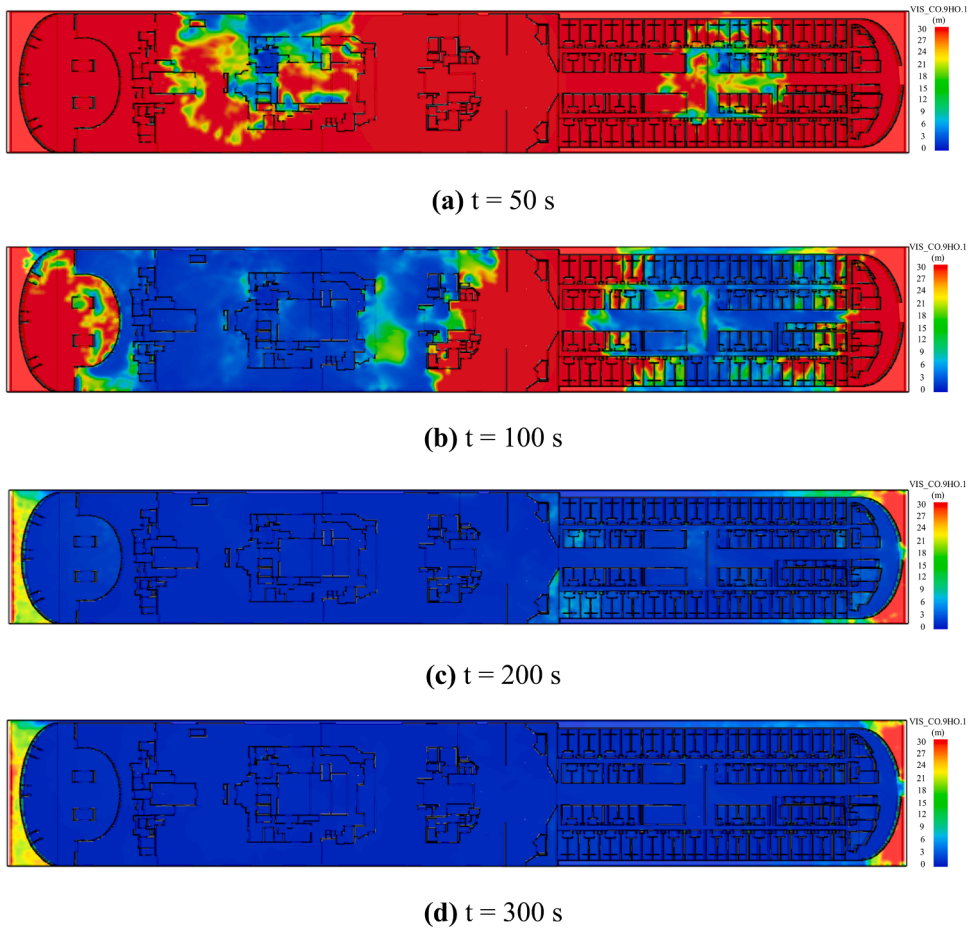


Fig. 18. Visibility at different times in case of a multi-source fire.

single-source fire. Since nodes 88, 89, 93, and 96 were inaccessible, the evacuation route for node 98 was 98→100→103→106→109→113→114, and the evacuation time was 162 s. Similarly, the evacuation route for node 80 was 80→78→76→73→74→71→70, with an evacuation time of 138 s.

The evacuation curves of both the TVEWDA and the Dijkstra algorithm using only distance between nodes are presented in Fig. 23.

Fig. 23 shows that both the TVEWDA and the Dijkstra algorithm using only distance between nodes exhibit similar trends. The evacuation time achieved by the TVEWDA is 171 s, while the Dijkstra algorithm using only distance between nodes records an evacuation time of 219 s. Notably, the TVEWDA outperforms the Dijkstra algorithm using only distance between nodes by reducing the evacuation time by 22%. This amounts to a time reduction of 48 s when considering emergency behavior and crowding. The effectiveness of the TVEWDA can be attributed to its consideration of fire-induced emergency behavior, which speeds up the evacuation process along the escape route, and its ability to intelligently avoid congestion.

3.3. Impact of the number of evacuees

In the case of a multi-source fire, the study investigated the impact of different numbers of evacuees on evacuation time. Simulations were conducted with 300, 400, 500, and 600 evacuees, and the corresponding results are shown in Fig. 24.

Fig. 24 shows that the TVEWDA exhibits shorter evacuation times compared to the Dijkstra algorithm using only distance between nodes. The evacuation times of both algorithms increase as the number of evacuees increases. When the total number of evacuees is 600, the TVEWDA achieves a 22% reduction in evacuation time compared to the Dijkstra algorithm using only distance between nodes.

3.4. Guided evacuation routes for evacuees

The TVEWDA is proposed to calculate the optimal evacuation route for evacuees. These routes are then introduced into a simulation platform. The simulation results are shown in Fig. 25. Li et al. [54] used the improved social force model within the software to simulate emergency behavior and congestion during passenger ship evacuations. The software employed in this study is AnyLogic,

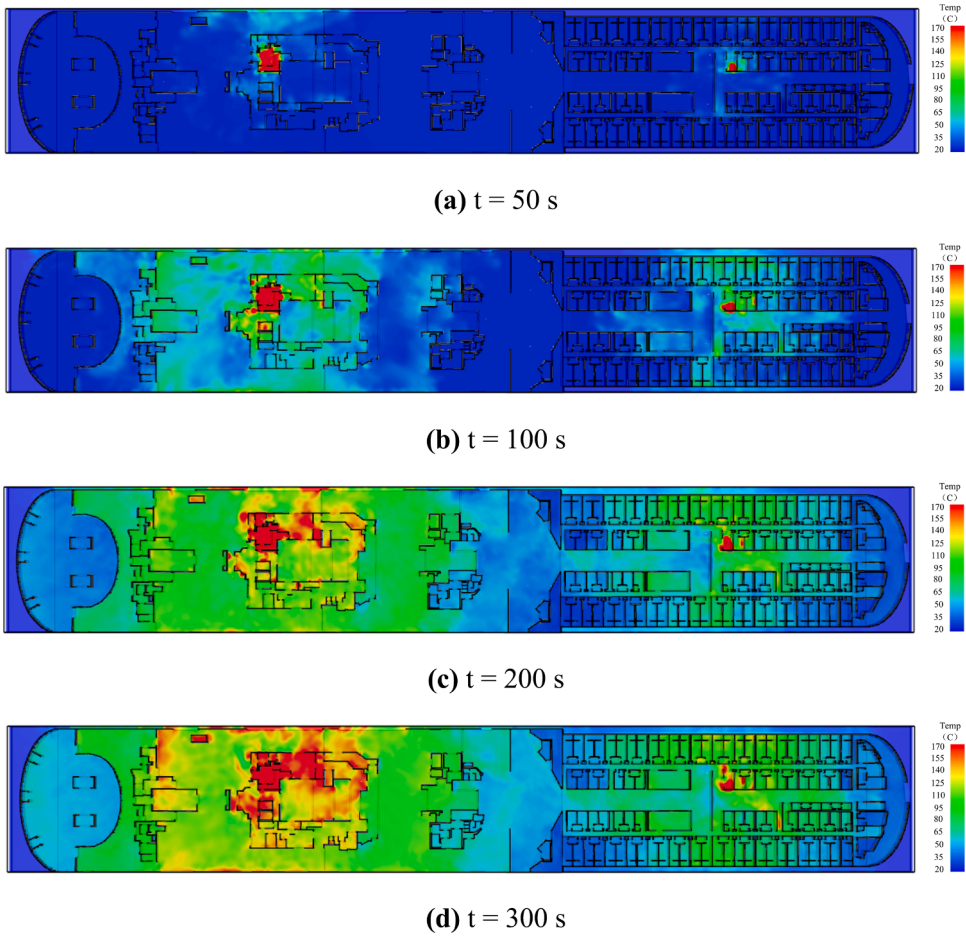


Fig. 19. Temperature at different times in case of a multi-source fire.

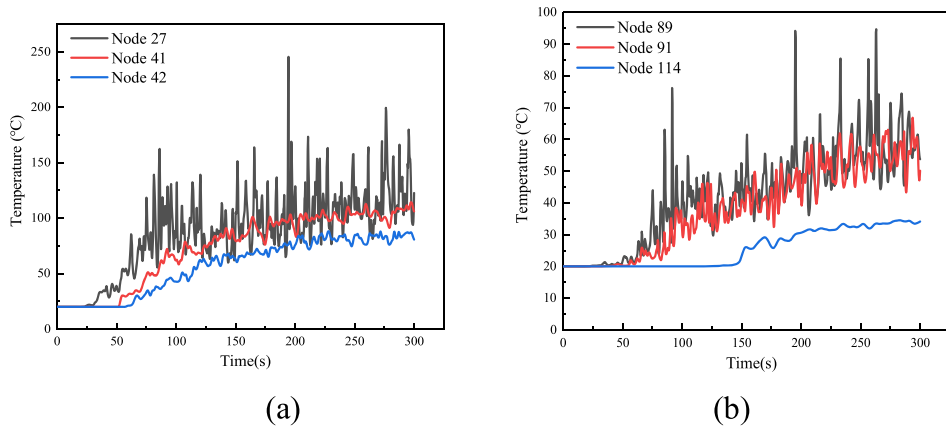


Fig. 20. Change in temperature at key locations.

which is widely used for passenger ship evacuation simulations, and it is based on the social force model. The improved social model of evacuation is given in (18)-(22) [55]:

$$m_a \frac{dv_a(t)}{dt} = m_a \frac{v'_a(t)e_a^0 - v_a(t)}{\tau_a} + \sum_{a \neq b} f_{ab} + \sum_w f_{aw}, \quad (18)$$

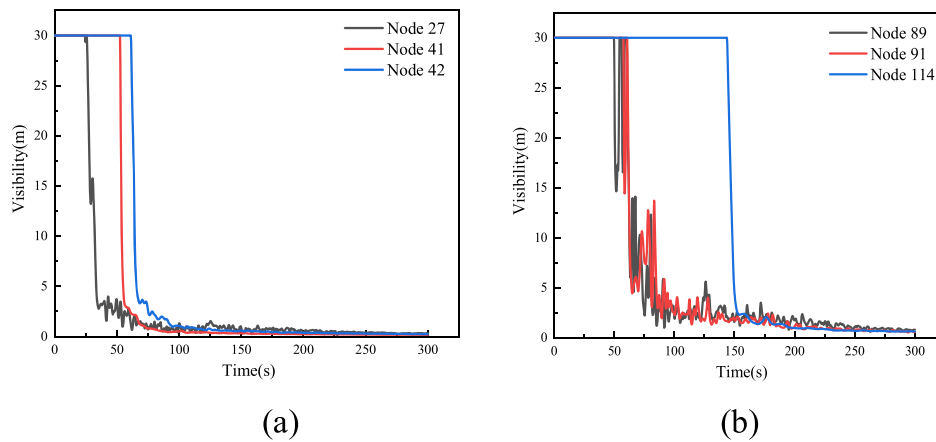


Fig. 21. Change in visibility at key locations.

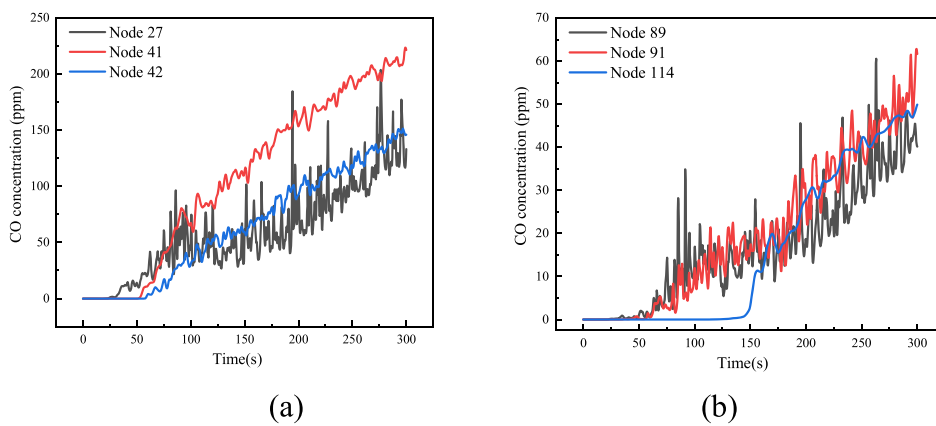


Fig. 22. Change in CO concentration at key locations.

Table 5

Evacuation routes for some evacuees in case of a multi-source fire ($t = 50$ s).

Start	End	Optimal route of the TVEWDA	Time(s)
38	42	38→40→39→43→42	33.7
47	42	47→44→43→42	24
96	91	96→93→94→95→92→91	135
98	114	98→100→103→106→109→113→114	146
104	114	104→107→110→115→114	40.2
106	114	106→109→113→114	39.6
107	114	107→110→115→114	28.9

Table 6

Evacuation routes for some people in case of a multi-source fire ($t = 100$ s).

Start	End	Optimal route of the TVEWDA	Time(s)
1	16	1→2→6→16	46.8
3	16	3→2→6→16	79
38	42	11→12→7→6→16	48.5
83	91	83→85→87→92→91	50.6
80	91	80→78→76→73→74→71→70	138
98	114	98→100→103→106→109→113→114	162
101	114	101→104→107→110→115→114	61.8
105	114	105→108→111→116→115→114	69.8

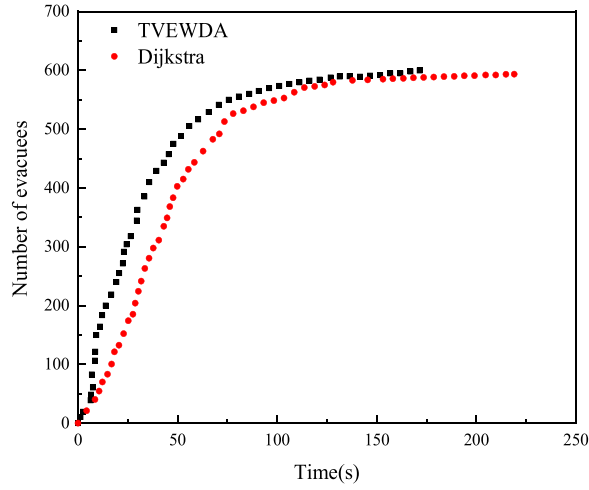


Fig. 23. Comparison of evacuation curves for different algorithms.

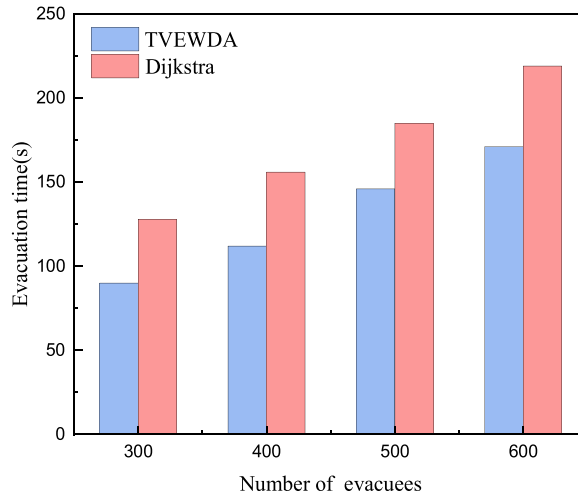


Fig. 24. Impact of the number of evacuees on evacuation time.



Fig. 25. Simulation results of the TVEWDA.

$$v'_a = 0.5\eta(E_r v_{\min} + \bar{v}_a), \quad (19)$$

$$\eta = \lambda \exp\left(\frac{-M}{P_{Et}}\right), \quad (20)$$

$$f_{ab} = A_a e^{[(r_{ab}-d_{ab})/B_a]} n_{ab} + k_0 g(r_{ab}-d_{ab}) n_{ab} + \nu g(r_{ab}-d_{ab}) \Delta v'_{ba} t_{ab}, \quad (21)$$

$$f_{aw} = A_w e^{[(r_a-d_{aw})/B_w]} n_{aw} + k_0 g(r_a-d_{aw}) n_{aw} + \nu g(r_a-d_{aw}) \Delta v'_{wa} t_{aw}, \quad (22)$$

Table 7

Parameter values in the improved social force model.

Symbol	Definition	Value
m	Mass of evacuee	64.3 kg
τ	Relaxation time	0.35 s
A_a	Intensity of social exclusion (evacuee to evacuee)	2000 N
B_a	Characteristic distance of social exclusion (evacuee to evacuee)	0.04 m
r_a	Radius of body	0.25 m
A_w	Intensity of social exclusion (evacuee to wall)	2000 N
B_w	Characteristic distance of social exclusion (evacuee to wall)	0.08 m
k_0	Extrusion coefficient.	120,000 kg s^{-2}
ν	Sliding friction coefficient	40,000 $\text{kg m}^{-1} \text{s}^{-1}$
g	The acceleration of gravity	9.8 m/s^2
λ	Adjustment factor	2
P_E	Individual's extroversion	1.5
M	Total number of evacuees	600
$\bar{v}_a$	The desired speed	1.8 m/s
$v_{\min}$	The minimum velocity	0.55 m/s
E_r	Constant	0.2

where m_a is the mass of evacuee a , $v_a(t)$ is the actual speed at time t , v_a^c is the corrected evacuation speed of a , e_a^0 is the direction of the desired speed of a , τ_a is the reaction time, η is the emergency behavior factor, E_r is a constant, $v_{\min}$ is the minimum velocity of evacuee, $\bar{v}_a$ is the desired speed, λ is the adjustment factor, M is total number of evacuees, P_E is individual's extroversion, r_{ab} is the sum of body radius of two evacuees a and b , d_{ab} is the distance between evacuees a and b , A_a is the intensity of social exclusion, B_a is the characteristic distance of social exclusion, n_{ab} is the unit vector of evacuee a pointing towards evacuee b , k_0 is the extrusion coefficient, ν is the sliding friction coefficient, Δv_{ab}^t is the change in speed, t_{ab} is the tangential direction, f_{ab} represents the interaction force, r_a is the positional coordinates of evacuee a , f_{aw} represents the wall force between evacuee a and obstacle w , d_{aw} is the distance between evacuee a and obstacle w , n_{aw} is the normalized vector, Δv_{wa}^t is the change in speed, A_w is the intensity of social exclusion (evacuee to wall), B_w is the characteristic distance of social exclusion (evacuee to wall).

Parameter values in the improved social force model are shown in Table 7.

A comparative analysis is also conducted in the current study using an improved modified social force model that considers both emergency behavior and congestion of evacuees. The simulation results obtained using this improved model are compared with those of the TVEWDA, as depicted in Fig. 26. The location and distribution of people, as well as the evacuation speed obtained with the improved model are the same as with the TVEWDA.

Fig. 25 and Fig. 26 illustrate the simulated evacuation routes of evacuees using both the TVEWDA and the improved social force model. In these figures, the darker colored areas in the diagram indicate higher concentrations of evacuees passing through the area.

The simulation results of the TVEWDA compared with the improved social force model are shown in Table 8.

The TVEWDA algorithm achieved an evacuation time of 170.5 s, while the improved social force model resulted in an evacuation time of 191.4 s. It is evident that the TVEWDA algorithm outperforms the improved social force model in terms of evacuation time. This is mainly because the TVEWDA considers the effect of fire and reasonably avoids the excessive congestion at certain staircases. The distribution of evacuees at each exit is shown in Table 9.

Table 9 presents a comparison of the two algorithms, highlighting their performance in terms of the number of evacuees at each exit. The TVEWDA demonstrates a more balanced distribution of evacuees across exits compared to the improved social force model.

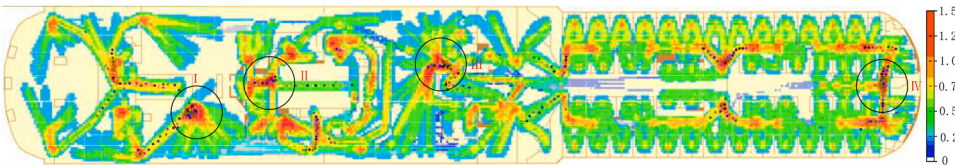
**(a)** Improved social force model**(b)** Passenger ship evacuation density map**Fig. 26.** Simulation results of the improved social force model.

Table 8
Simulation results comparison (s).

Parameters	TVEWDA	Improved social force model
Evacuation time	170.5	191.4
Evacuation time standard deviation	39.0	38.3
Average evacuation time	69.1	73.8

Table 9
Number of evacuees at each exit on the fifth deck.

Algorithm	Number of evacuees (per)										
	Exit 16	Exit 18	Exit27	Exit 41	Exit 42	Exit 58	Exit 70	Exit 72	Exit 89	Exit 91	Exit 114
Improved social force model	48	50	46	56	52	71	29	34	62	56	96
TVEWDA	58	50	41	62	57	65	32	41	64	52	78

Table 10
Congestion time of the two algorithms.

Congested area	TVEWDA (s)	Improved social force model (s)
I	18	22
II	32	36
III	25	34
IV	33	46

Specifically, at exit 114, the improved social force model records 96 evacuees, while TVEWDA shows 78 evacuees. In terms of the minimum number of evacuees, the improved social force model indicates 29 evacuees at exit 70, whereas the TVEWDA indicates 32 evacuees at the same exit. Overall, the TVEWDA outperforms the improved social force model in terms of route planning for cruise ship fire spread hazards.

During the fire evacuation, the congestion areas I, II, III and IV for evacuating passengers from passenger ships are shown in Fig. 26 (b). The congestion time of the two algorithms is presented in Table 10.

As indicated in Table 10, the congestion time of TVEWDA is consistently lower than that of the improved social force model in all four congested areas, resulting in a total evacuation time that is 20.9 s shorter for TVEWDA compared to the improved social force model.

4. Conclusion

This research proposes the TVEWDA based on the Dijkstra algorithm and FDS simulation. It determines the optimal evacuation route by considering real-time fire spreading. This modified algorithm also considers key fire risk factors, such as CO concentration, visibility, and temperature. Additionally, it introduces emergency behavior and congestion factors that affect the evacuees' movement speed, making the evacuation model more representative of real-world scenarios. The case studies conducted for single-source and multi-source fires show that the TVEWDA can dynamically adjust the emergency evacuation plan, provide decision-makers with optimal routes, and guide evacuees to evacuate promptly.

However, the current approach has certain limitations. For instance, it does not consider the influence of group behavior and the presence of guides during the evacuation process. Furthermore, recent studies have shown that the heeling angle of the cruise ship during the evacuation process also significantly impacts the evacuation. Therefore, future studies will aim at incorporating group behavior, the role of guides, and cruise ship heeling to improve the accuracy of evacuation route optimization of the event of fires.

CRediT authorship contribution statement

Linfan Liu: Conceptualization, Formal analysis, Investigation, Project administration, Software, Supervision, Validation. **Huajun Zhang:** Conceptualization, Formal analysis, Investigation, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing. **Jun Shi:** Conceptualization, Formal analysis, Investigation, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing. **Jianning Geng:** Conceptualization, Formal analysis, Investigation, Project administration, Software, Supervision, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests.

Data availability

Data will be made available on request.

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References

- [1] T. Browne, B. Veitch, R. Taylor, J. Smith, D. Smith, F. Khan, Consequence modeling for Arctic ship evacuations using expert knowledge, *Mar. Policy* 130 (2021), 104582, <https://doi.org/10.1016/j.marpol.2021.104582>.
- [2] M. Choi, S. Chi, Optimal route selection model for fire evacuations based on hazard prediction data, *Simul. Model. Pract. Theory* 94 (2019) 321–333, <https://doi.org/10.1016/j.simpat.2019.04.002>.
- [3] H. Kim, S. Han, Crowd evacuation simulation using active route choice model based on human characteristics, *Simul. Model. Pract. Theory* 87 (2018) 369–378, <https://doi.org/10.1016/j.simpat.2018.07.014>.
- [4] Y. Zang, Q. Mei, S. Liu, Evacuation simulation of a high-rise teaching building considering the influence of obstacles, *Simul. Model. Pract. Theory* 112 (2021), 102354, <https://doi.org/10.1016/j.simpat.2021.102354>.
- [5] H. Arshad, J. Emblemsvåg, G. Li, R. Ostnes, Determinants, methods, and solutions of evacuation models for passenger ships: a systematic literature review, *Ocean Eng.* 263 (2022), 112371, <https://doi.org/10.1016/j.oceaneng.2022.112371>.
- [6] P.A. Sarvari, E. Cevikcan, A. Ustundag, M. Celik, Studies on emergency evacuation management for maritime transportation, *Marit. Policy Manag.* 45 (5) (2018) 622–648, <https://doi.org/10.1080/03088839.2017.1407044>.
- [7] D. Lee, H. Kim, J.H. Park, B.J. Park, The current status and future issues in human evacuation from ships, *Safety Sci.* 41 (10) (2003) 861–876, [https://doi.org/10.1016/S0925-7535\(02\)00046-2](https://doi.org/10.1016/S0925-7535(02)00046-2).
- [8] E. Çakır, Determinants of medical evacuations from merchant cargo ships: evidence from telemedical assistance service of Turkey data, *Ocean Coast. Manag.* 211 (2021), 105797, <https://doi.org/10.1016/j.ocecoaman.2021.105797>.
- [9] M. Hu, W. Cai, Research on the evacuation characteristics of cruise ship passengers in multi-scenarios, *Appl. Sci.* 12 (9) (2022) 4213, <https://doi.org/10.3390/app12094213>, Basel.
- [10] Z. Liu, Y. Li, Z. Zhang, W. Yu, A new evacuation accessibility analysis approach based on spatial information, *Reliab. Eng. Syst. Safe* 222 (2022), 108395, <https://doi.org/10.1016/j.res.2022.108395>.
- [11] G. Vukelić, G. Vizenin, V. Francić, Prospects for use of extended reality technology for ship passenger evacuation simulation, *Pomorstvo* 35 (1) (2021) 49–56, <https://doi.org/10.31217/p.35.1.6>.
- [12] J. Li, W. Liu, F. Zhang, T. Li, R. Wang, A ship fire escape speed correction method considering the influence of crowd interaction, *Mathematics* 10 (15) (2022) 2749, <https://doi.org/10.3390/math10152749>.
- [13] G. González-Cela, R. Bellas, R. Carreño, J. Martínez, R. Touza, A. González-Gil, D. González, A framework for integrating human factors in the early stages of ship design: application to the mess halls of a surface combatant, *J. Sh. Prod. Des.* 36 (01) (2020) 1–13, <https://doi.org/10.5957/jspd.2020.36.1.1>.
- [14] X. Wang, Z. Liu, S. Loughney, Z. Yang, Y. Wang, J. Wang, Numerical analysis and staircase layout optimisation for a Ro-Ro passenger ship during emergency evacuation, *Reliab. Eng. Syst. Safe* 217 (2021), 108056, <https://doi.org/10.1016/j.res.2021.108056>.
- [15] A.I. Ginnis, K.V. Kostas, C.G. Politis, P.D. Kaklis, VELOS: a VR platform for ship-evacuation analysis, *Comput. Aided Des.* 42 (11) (2010) 1045–1058, <https://doi.org/10.1016/j.cad.2009.09.001>.
- [16] C.W. Chu, H.A. Lu, C.Z. Pan, Emergency evacuation route for the passenger ship, *J. Mar. Sci. Technol.* 21 (5) (2021) 3, <https://doi.org/10.6119/JMST-012-0529-3>.
- [17] P. Wang, T. Zhang, Y. Xiao, Emergency evacuation path planning of passenger ship based on cellular ant optimization model, *J. Shanghai Jiaotong Univ.* 25 (6) (2020) 721–726, <https://doi.org/10.1007/s12204-020-2215-y>.
- [18] D. Łozowicka, The design of the arrangement of evacuation routes on a passenger ship using the method of genetic algorithms, *PLOS One* 16 (8) (2021), e0255993, <https://doi.org/10.1371/journal.pone.0255993>.
- [19] L. Liu, H. Zhang, Y. Zhan, Y. Su, C. Zhang, Intelligent optimization method for the evacuation routes of dense crowds on cruise ships, *Simul. Model. Pract. Theory* 117 (2022), 102496, <https://doi.org/10.1016/j.simpat.2022.102496>.
- [20] K. Sriniketh, A.V. Le, R.E. Mohan, B.J. Sheu, V.D. Tung, P. Van Duc, M.B. Vu, Robot-aided human evacuation optimal path planning for fire drill in buildings, *J. Build. Eng.* 72 (2023), 106512, <https://doi.org/10.1016/j.job.2023.106512>.
- [21] F.T. Yan, Y.H. Hu, J.Y. Jia, Q.H. Guo, H.H. Zhu, Z.G. Pan, RFES: a real-time fire evacuation system for Mobile Web3D, *Front. Inf. Technol. Electron. Eng.* 20 (8) (2019) 1061–1074, <https://doi.org/10.1631/FITEE.1700548>.
- [22] A.F. Lee, P.H. Tsai, Online indoor fire evacuation system, *IEEE Syst. J.* (2023), <https://doi.org/10.1109/JSYST.2023.3238348>.
- [23] G. Bernardini, M. Azzolini, M. D'Orazio, E. Quagliarini, Intelligent evacuation guidance systems for improving fire safety of Italian-style historical theatres without altering their architectural characteristics, *J. Cult. Herit.* 22 (2016) 1006–1018, <https://doi.org/10.1016/j.culher.2016.06.008>.
- [24] S. Cao, L. Fu, W. Song, Exit selection and pedestrian movement in a room with two exits under fire emergency, *Appl. Math. Comput.* 332 (2018) 136–147, <https://doi.org/10.1016/j.amc.2018.03.048>.
- [25] X. Yang, R. Zhang, Y. Li, F. Pan, Passenger evacuation path planning in subway station under multiple fires based on multi-objective robust optimization, *IEEE Trans. Intell. Transp. Syst.* 23 (11) (2022) 21915–21931, <https://doi.org/10.1109/TITS.2022.3190291>.
- [26] Y. Ma, K. Liu, M. Chen, J. Ma, X. Zeng, K. Wang, C. Liu, ANT: deadline-aware adaptive emergency navigation strategy for dynamic hazardous ship evacuation with wireless sensor networks, *IEEE Access* 8 (2020) 135758–135769, <https://doi.org/10.1109/ACCESS.2020.3011545>.
- [27] I. Kim, H. Kim, D. Chang, D.H. Jung, H.G. Sung, S.K. Park, B.C. Choi, Emergency evacuation simulation of a floating LNG bunkering terminal considering the interaction between evacuees and CFD data, *Safety Sci.* 140 (2021), 105297, <https://doi.org/10.1016/j.ssci.2021.105297>.
- [28] K. Deng, Q. Zhang, H. Zhang, P. Xiao, J. Chen, Optimal emergency evacuation route planning model based on fire prediction data, *Mathematics* 10 (17) (2022) 3146, <https://doi.org/10.3390/math10173146>.
- [29] J. Ji, Z. Ma, J. He, Y. Xu, Z. Liu, Research on risk Evaluation and dynamic escape path planning algorithm based on real-time spread of ship comprehensive fire, *J. Mar. Sci. Eng.* 8 (8) (2020) 602, <https://doi.org/10.3390/jmse8080602>, 2020.
- [30] Y. Yuan, D. Wang, Path selection model and algorithm for emergency logistics management, *Comput. Ind. Eng.* 56 (2009) 1081–1094, <https://doi.org/10.1016/j.cie.2008.09.033>.
- [31] K.J. Spyrou, I.A. Koromila, A risk model of passenger ship fire safety and its application, *Reliab. Eng. Syst. Safe* 200 (2020), 106937, <https://doi.org/10.1016/j.res.2020.106937>.
- [32] S. Fang, Z. Liu, S. Zhang, X. Wang, Y. Wang, S. Ni, Evacuation simulation of an Ro-Ro passenger ship considering the effects of inclination and crew's guidance, *J. Eng. Marit. Environ.* 237 (1) (2023) 192–205, <https://doi.org/10.1177/14750902221106566>.
- [33] C.T. Ng, T.C.E. Cheng, E. Levner, B. Kriheli, Optimal bi-criterion planning of rescue and evacuation operations for marine accidents using an iterative scheduling algorithm, *Ann. Oper. Res.* 296 (2021) 407–420, <https://doi.org/10.1007/s10479-020-03632-6>.

- [34] J. Li, H. Zhu, A risk-based model of evacuation route optimization under fire, *Procedia Eng.* 211 (2018) 365–371, <https://doi.org/10.1016/j.proeng.2017.12.024>.
- [35] F. Mirahadi, B.Y. McCabe, EvacuSafe: a real-time model for building evacuation based on Dijkstra's algorithm, *J. Build. Eng.* 34 (2021), 101687, <https://doi.org/10.1016/j.jobe.2020.101687>.
- [36] Y. Zhu, H. Li, Z. Wang, Q. Li, Z. Dou, W. Xie, W. Nie, Optimal evacuation route planning of urban personnel at different risk levels of flood disasters based on the improved 3D Dijkstra's Algorithm, *Sustainability* 14 (16) (2022) 10250, <https://doi.org/10.3390/su141610250>.
- [37] Y. Darmi, M. Muntahanah, Implementation of Dijkstra's algorithm to determine evacuation paths in Earthquake and Tsunami simulations in Bengkulu city based on Gis, *J. Media Comput. Sci.* 1 (2) (2022) 117–130, <https://doi.org/10.37676/jmcs.v1i2.2812>, 117–130.
- [38] I.P. Sari, M.F. Fahrenza, M.I. Mufit, I.F. Qathrunad, Implementation of Dijkstra's algorithm to determine the shortest route in a city, *J. Comput. Sci. Inf. Technol. Telecommun. Eng.* 2 (1) (2021) 134–138, <https://doi.org/10.30596/jcositte.v2i1.6503>.
- [39] PyroSim User Manual, Thunderhead Support (thunderheadeng.com) (accessed on June 2022).
- [40] T. Korhonen, S. Hostikka, Fire dynamics simulator with evacuation FDS + Evac. Technical reference and user's guide, VTT Technical Research Centre of Finland, 2009, September 10, 2009. (accessed on June 2022).
- [41] K.B. McGrattan, G.P. Forney, J. Floyd, S. Hostikka, K. Prasad, Fire Dynamics Simulator (version 4)—User's Guide, US Department of Commerce, Technology Administration, National Institute of Standards and Technology, 2005.
- [42] R.L. Hansen, G.G. Back, Water spray protection of machinery spaces, *Fire Technol.* 37 (2001) 317–326.
- [43] G. Fang, R.D. Susanto, S. Wirasantosa, F. Qiao, A. Supangat, B. Fan, S. Li, Volume, heat, and freshwater transports from the South China Sea to Indonesian seas in the boreal winter of 2007–2008, *J. Geophys. Res. Ocean.* 115 (C12) (2010), <https://doi.org/10.1029/2010JC006225>.
- [44] Q. Xie, S. Guo, Y. Zhang, C. Wang, C. Ma, Q. Li, An integrated method for assessing passenger evacuation performance in ship fires, *Ocean Eng.* 262 (2022), 112256, <https://doi.org/10.1016/j.oceaneng.2022.112256>.
- [45] L. Biao, Research On Optimization and Application of Emergency Evacuation Paths Under Offshore Platform Fire Scenarios, China University of Petroleum (East China), 2019.
- [46] IMO. IMO Amendments to the Guidelines on Alternative Design and Arrangements for Fire Safety(MSC/Circ.1002), London, 2016 (accessed on June 2022).PDF (e-lass.eu).
- [47] S. Zhu, Study On a Composite Occupant Evacuation Simulation in Buildings Based On a Two-Zone Fire Model, Central South University, Changsha, China, 2010 in Chinese.
- [48] T. Jin, T. Yamada, Irritating effects of fire smoke on visibility, *Fire Sci. Technol.* 5 (1) (1985) 79–90.
- [49] J.A. Milke, Evaluating the early development of smoke hazard from fires in large spaces, *Build. Eng.* 106 (1) (2000) 627–636.
- [50] J.H. Wang, Research On the Uncertainties of Large-Scale Crowd Evacuation Under Unconventional Emergencies, University of Science and Technology of China, Hefei, 2013.
- [51] T. Tsunemine, E. Kadokawa, Y. Ueda, J. Fukumoto, T. Wada, K. Ohtsuki, H. Okada, Emergency urgent communications for searching evacuation route in a local disaster, in: Proceedings of the 5th IEEE Consumer Communications and Networking Conference, IEEE, 2013, pp. 1196–1200, <https://doi.org/10.1109/ccnc08.2007.267>.
- [52] A. Mohd Ibrahim, I. Venkat, P. De Wilde, M.R. Mohd Romlay, A. Bahamid, The role of crowd behavior and cooperation strategies during evacuation, *Simulation* 98 (2022) 737–751, <https://doi.org/10.1177/00375497221075611>.
- [53] S. Fang, Z. Liu, X. Yang, X. Wang, J. Wang, Z. Yang, A quantitative study of the factors influencing human evacuation from ships, *Ocean Eng.* 285 (2023), 115156, <https://doi.org/10.1016/j.oceaneng.2023.115156>.
- [54] J. Li, F. Zhang, M. Li, J. Li, An evacuation time correction method for passenger ships fire considering passenger panic, *IEEE Access* 10 (2022) 112433–112443, <https://doi.org/10.1109/ACCESS.2022.3216962>.
- [55] X. Yang, X. Yang, Y. Li, J. Zhang, Y. Kang, Obstacle avoidance in the improved social force model based on ant colony optimization during pedestrian evacuation, *Phys. A* 583 (2021), 126256, <https://doi.org/10.1016/j.physa.2021.126256>.